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INTRODUCTION

The North Atlantic right whale (NARW) is a critically endangered species of baleen whale, struggling to recover from whaling due to ship strikes and entanglement. During the winter, a portion of the population migrate from northern foraging grounds to give birth and nurse newborn calves off the coast of South Carolina, Georgia, and Florida. In the Southeast US (SEUS), a designated critical habitat zone requires vessels >65 ft to travel at speeds less than 10 knots from November 15-April 15. However, right whales are still at risk of a ship strike within this zone due to low compliance with speed regulations is low in the SEUS. Passive acoustic monitoring (PAM) is a common method for detecting the presence of baleen whales, and glider-based PAM is a useful tool in supplementing aerial-based detections by providing coverage when aerial surveillance is not feasible due to weather or logistical constraints. Beginning in 2023, we launched a pilot PAM program using a glider and bottom mounted hydrophones in the SEUS. The glider was deployed for a 2-week mission in January 2023, two 4-week missions in January through March 2024, and one 4-week mission and one 6 day mission in January and February 2025. During these missions, there were 16 possible and 3 definite right whale acoustic detections. A detection on January 20, 2024 was the first definite glider-based acoustic detection in the Southeast US and communicated in real-time via WhaleMap, Whale Alert and Robots4Whales and the Early Warning System notifications program.

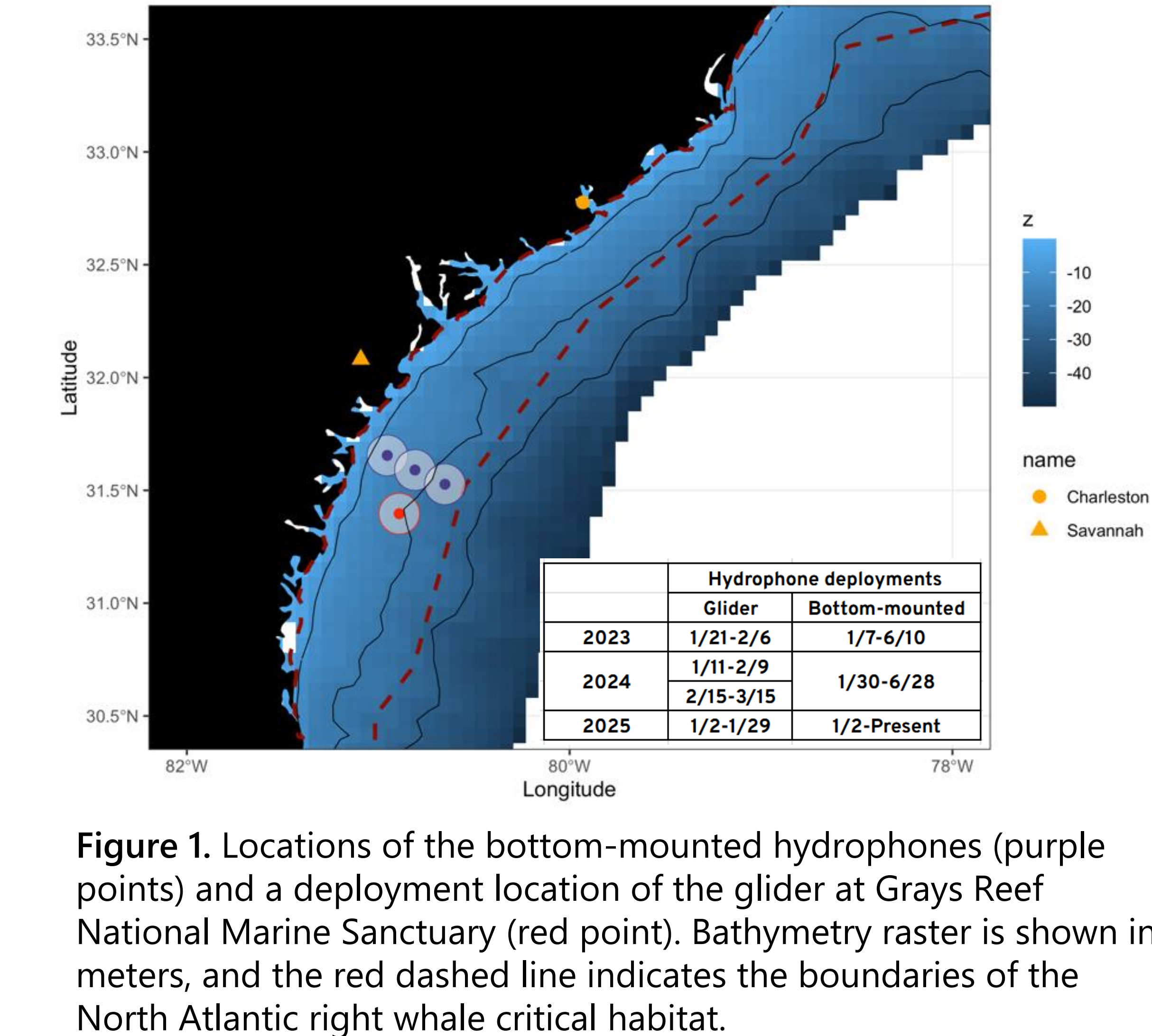


Figure 1. Locations of the bottom-mounted hydrophones (purple points) and a deployment location of the glider at Grays Reef National Marine Sanctuary (red point). Bathymetry raster is shown in meters, and the red dashed line indicates the boundaries of the North Atlantic right whale critical habitat.

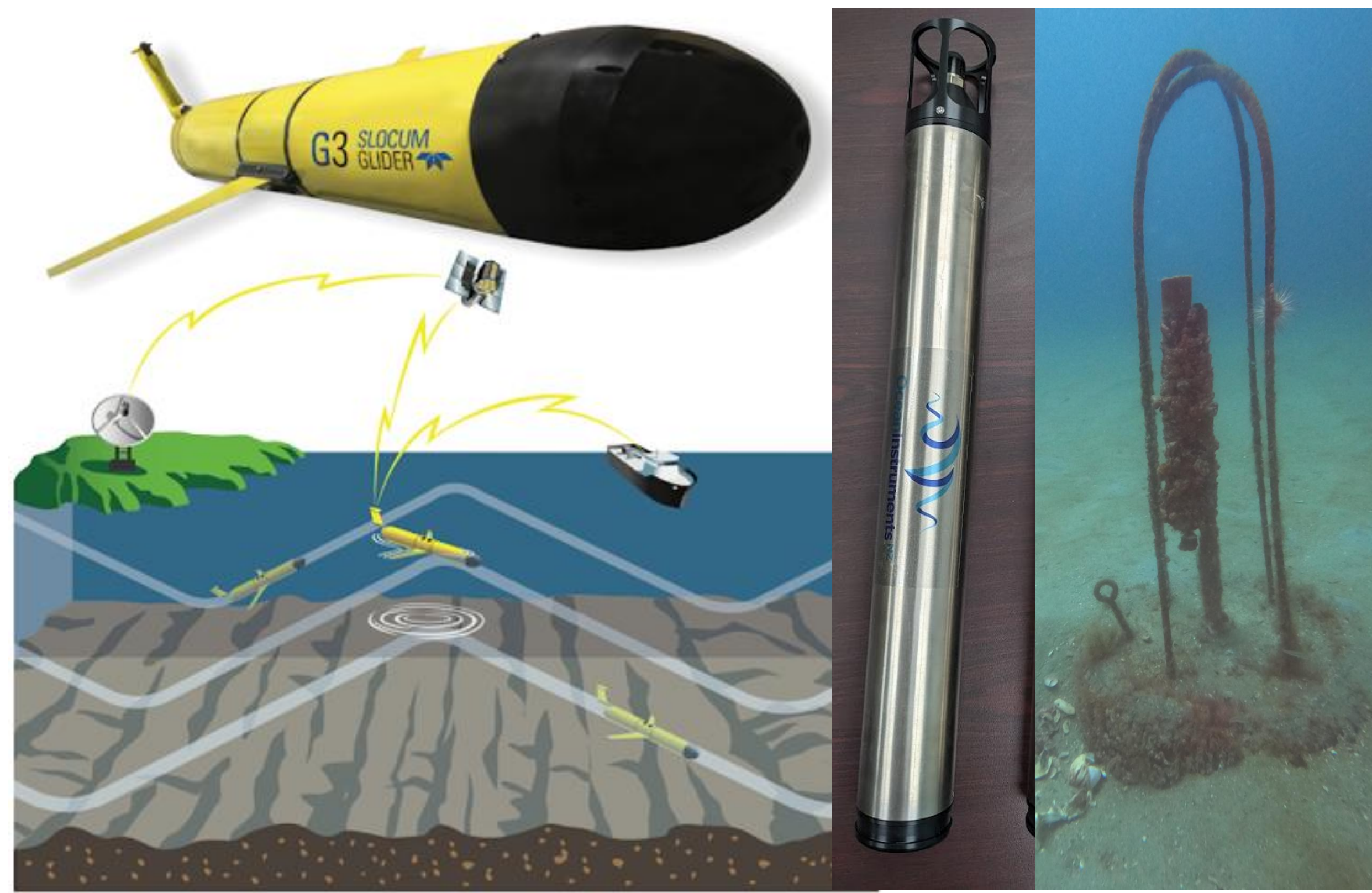


Figure 2. (above) Left: Slocum G3 Glider. Bottom: Schematic of yo-yo flight pattern and communications via cell phone and satellite Iridium link. Right: ST600 from OceanInstruments and the Sound Trap deployment configuration of our moorings upon recovery in 2023.

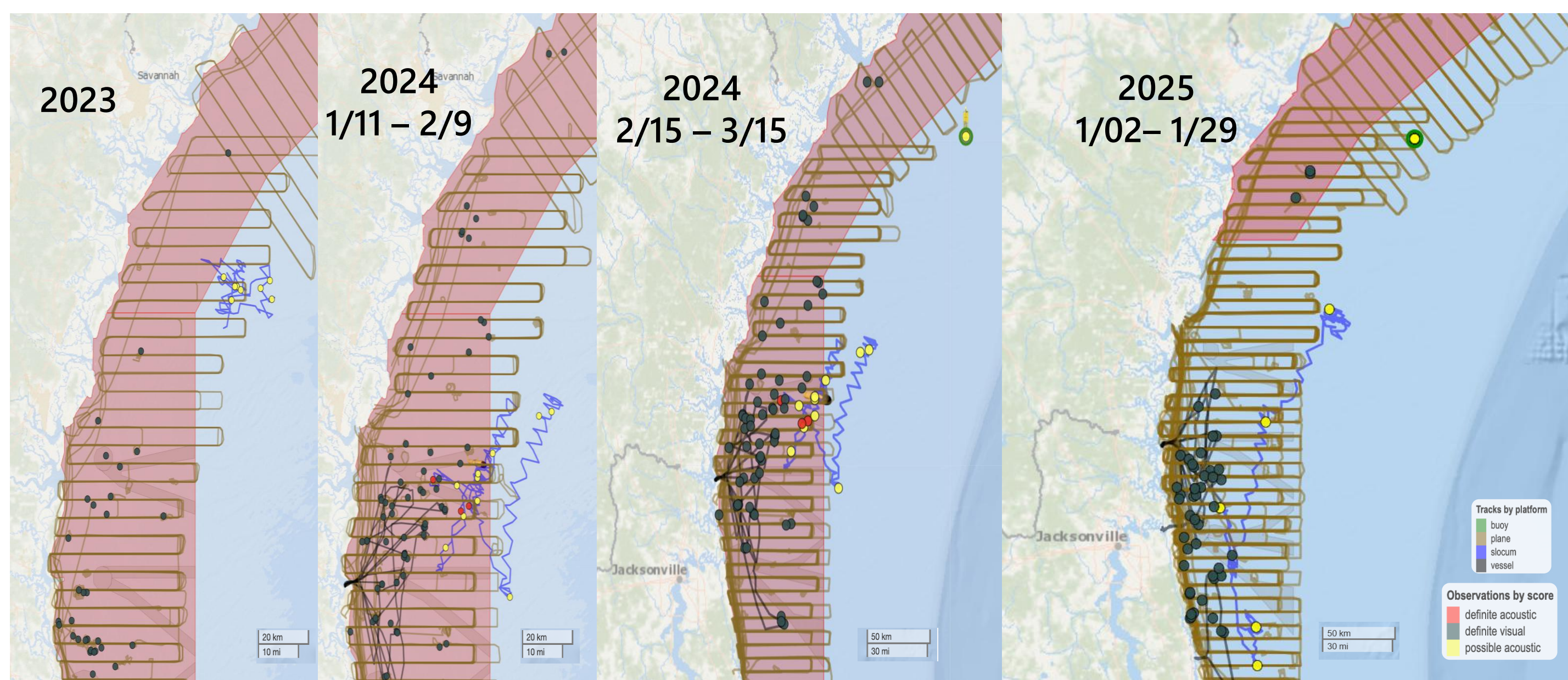


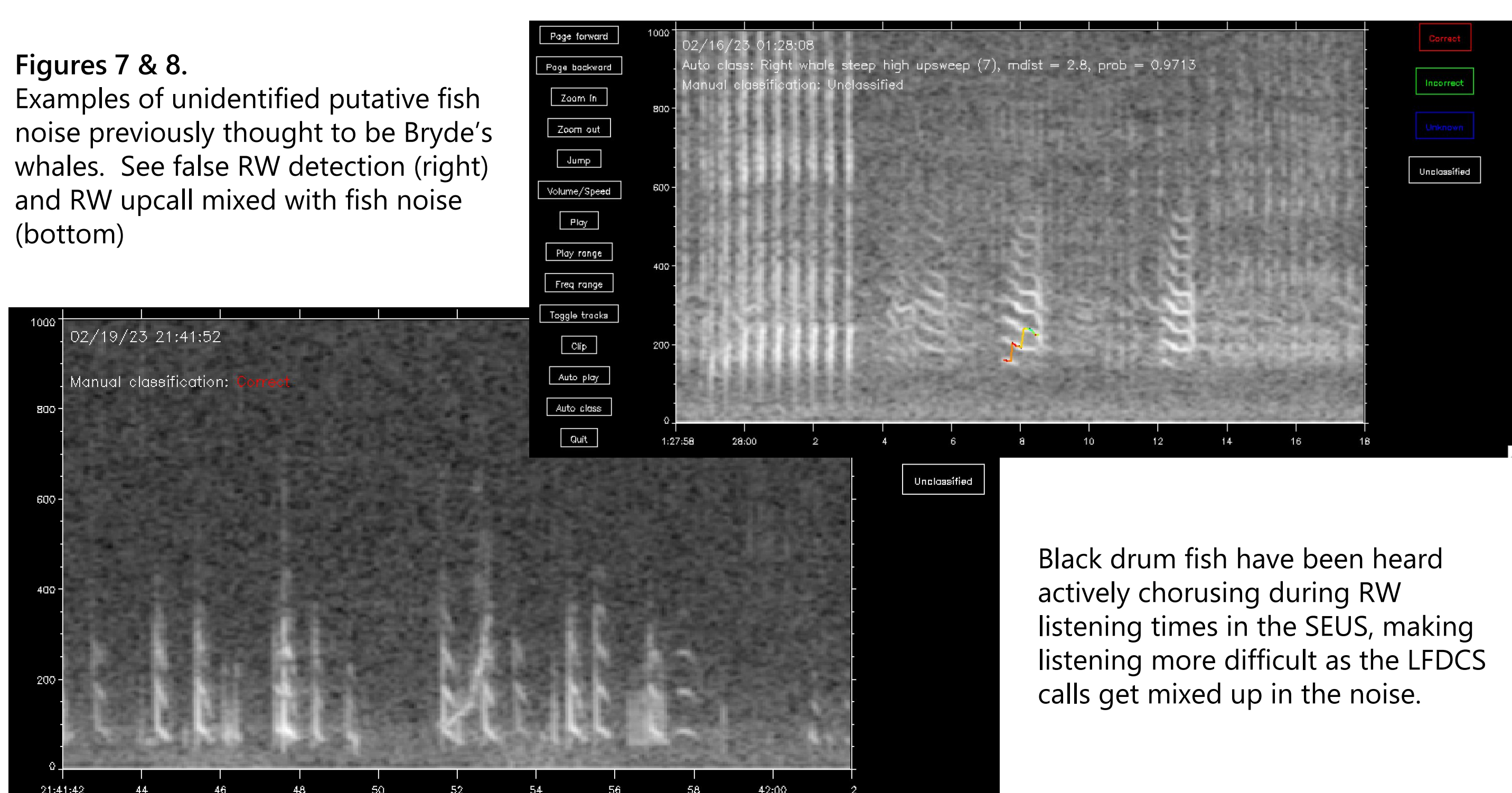
Figure 4: Glider paths are depicted in blue. The red polygon is the right whale SMA. Acoustic detections are marked in red, possible acoustic detections in yellow, and visual detections in green. Aerial and vessel survey effort is shown in light brown and dark brown respectively. Maps from WhaleMap.

OTHER SOUNDS

- The classification software filters broadband sounds and highlights sounds that could be produced by baleen whales. It classifies pitch tracks by species: right whale, fin whale, sei whale, and humpback whale.
- The second mission of 2024, February-March, the real-time analysis detected fin whales on 27 out of 30 days; however, the locations were outside of their typical range.
- The pitch tracks in the real-time analysis met the required criteria for detection, but further analysis with spectrograms of archival data show the detections were incorrect caused by glider self-noise.
- Additionally, fish choruses unique to the SEUS during right whale calving season produce pitch tracks in the real-time analysis.
- The loud soundscape contributes to high data usage over short time periods, and potential masking of right whale calls.

Figures 7 & 8.

Examples of unidentified putative fish noise previously thought to be Bryde's whales. See false RW detection (right) and RW upcall mixed with fish noise (bottom)



Black drum fish have been heard actively chorusing during RW listening times in the SEUS, making listening more difficult as the LFDCS calls get mixed up in the noise.

Figures 5 & 6. Example of a real fin whale call (top) versus glider noise (bottom) mistakenly detected as a fin whale on the LFDCS

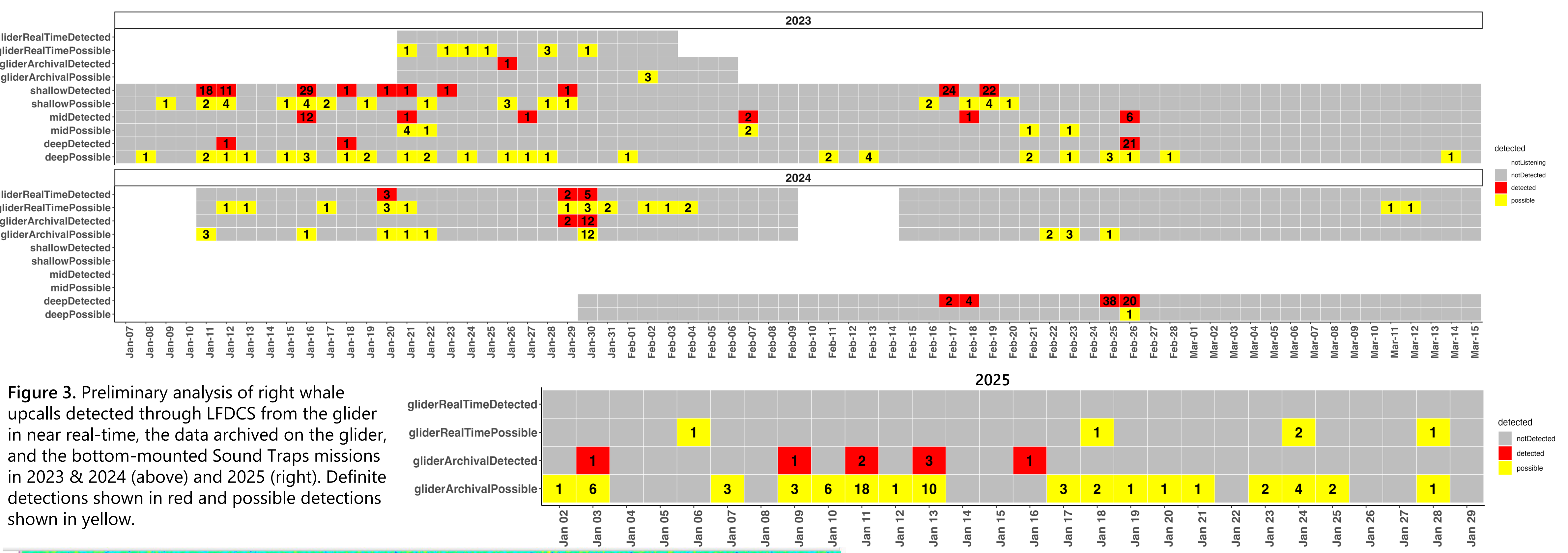
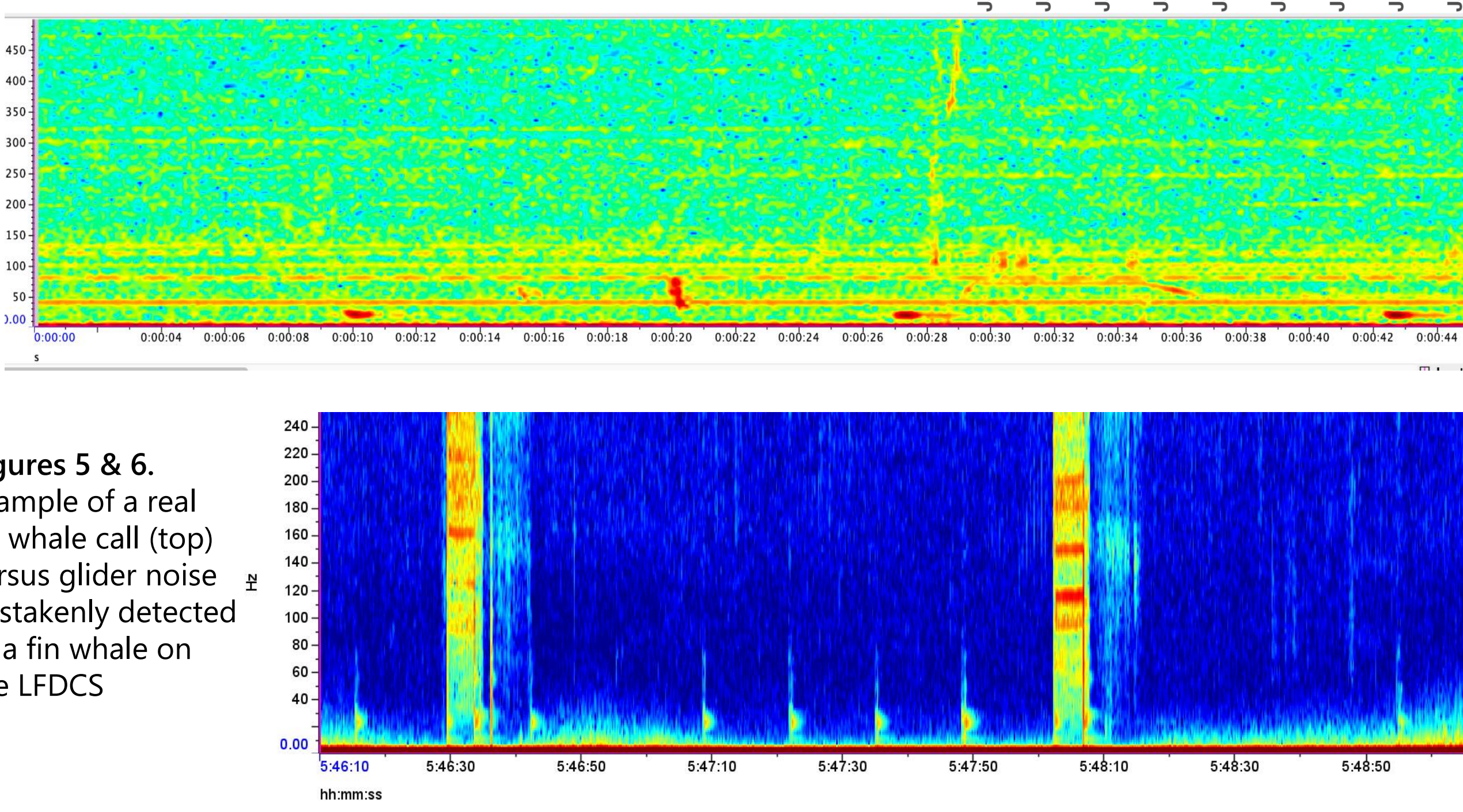


Figure 3. Preliminary analysis of right whale upcalls detected through LFDCS from the glider in near real-time, the data archived on the glider, and the bottom-mounted Sound Traps missions in 2023 & 2024 (above) and 2025 (right). Definite detections shown in red and possible detections shown in yellow.



METHODS

A Slocum G3 Glider equipped with a digital acoustic monitoring instrument (DMON2) was deployed for 3 missions off the coast of Georgia (Fig. 1 inset; Fig. 4). The glider was programmed to surface at 4 hour intervals to enable near real-time baleen whale vocalization detections (Fig 2). Three Ocean Instruments Sound Traps (ST600) were used as bottom-mounted, archival hydrophones and deployed at increasing depths (15.9m, 19.5m, 22m) and increasing distances off the coast of Savannah, Georgia (Fig. 1, Fig. 2). The low-frequency detection and classification system (LFDCS) was used to identify and classify calls from the real-time glider acoustic data as well as the archival hydrophone and archival glider. Detections of right whale vocalizations during the deployment period were compared between each of the three bottom-mounted hydrophones and the glider to inform optimal PAM deployment depth to detect right whales in this region.

RESULTS

Right whales were detected in near real-time on 3 days and possibly detected on 20 days with the glider across all 4 missions. Additional up-calls were observed on the glider and bottom-mounted archival data (Fig. 3 & 4). Definite right whale detections were made on all bottom mounted hydrophones in 2023 (Fig. 3), with majority occurring before and after the glider was operational. Bottom mounted hydrophones in 2024 were either not recovered (shallow), corrupted (mid) or made detections (deep). The archival data for 2025 has not been recovered yet, but definite detections were made on the archival glider data. Archival analysis has shown loud communities of fish that could reduce the ability to detect faint right whale calls and produce false detections in LFDCS.

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